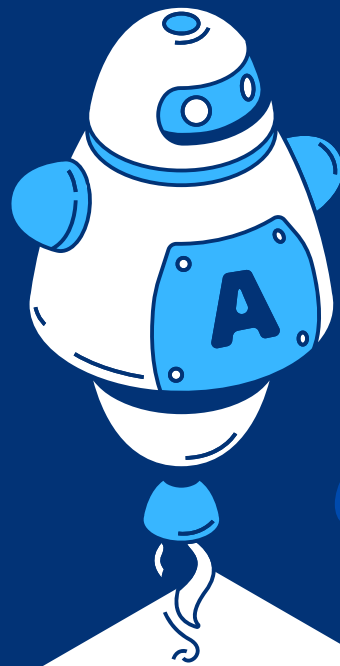




GENQUANTIS
PVT. LTD.



DATA SCIENCE COURSE BY ACADEMY GENQUANTIS

For More Information, visit our website: academy.genquantis.com or reach out to us at:
academy@genquantis.com, ph no: 7036955133

Know your future as data scientist & artificial intelligence professional

YOU CAN APPLY FOR

Data Scientist

Machine Learning Engineer

Deep Learning Engineer

Data Mining Specialist

AI/ML Model Validator

AI Search Scientist

Data Analyst

Data Visualisation Specialist

AI Product Manager

AI Chatbot Designer

Big Data Engineer

AI Consultant

Quantitative Analyst

AIOps Specialist

NLP Engineer

Computer Vision Engineer

Business Intelligence Analyst

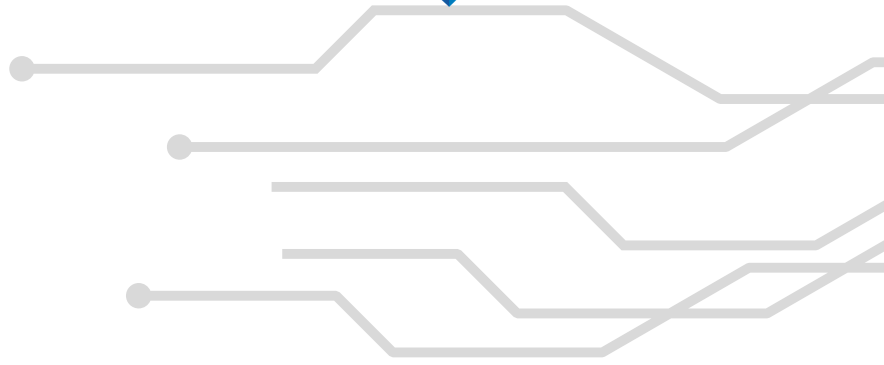
Algorithm Developer

Conversational AI Developer

AI Solution Architect

Data Engineering

Skillset Covered



Data Analytics /
Business Analytics



Ensemble
Techniques



Data Visualisation



Natural Learning
Processing



Machine Learning
Algorithms



Forecasting
Analytics



Statistics



Generative
AI

Learning path flow





Course Module Information

Module 1: Python Programming Basics

1. Python Setup & Environment configurations management guidelines and IDE selection process
2. Variables & Data Types including strings, integers, floats, booleans and type conversions
3. Lists, Tuples, Dictionaries with indexing, slicing operations and nested structure usage
4. Conditional Statements using if, elif, else blocks with logical operator implementations
5. Loops including for and while iterations with range functions and control flow statements
6. Functions & Lambda expressions with parameters, returns, scope and closures understanding

Module 2: Python for Data Analysis

1. NumPy Basics including array creation, indexing, slicing operations and vectorized computations for numerical data processing
2. Pandas DataFrames covering data structure creation, column operations, indexing, and basic data manipulation techniques
3. Reading CSV, Excel, JSON files with file handling methods and loading structured data into DataFrames
4. Data Filtering & Selection using conditional operations, boolean indexing and subset extraction techniques
5. GroupBy & Aggregations for summarizing datasets with functions like sum, mean, count and multi-level grouping
6. Handling Missing Data using detection, removal, imputation techniques and data cleaning best practices

Module 3: Statistics for Data Science

1. Mean, Median, Mode including measures of central tendency and their application in summarizing datasets
2. Variance & Standard Deviation covering data dispersion, spread analysis and variability measurement techniques
3. Probability Basics including fundamental concepts, rules, events and likelihood estimation in data scenarios
4. Distributions such as Normal and Binomial with properties, visualization and real-world data modeling applications
5. Hypothesis Testing covering null and alternative hypotheses, p-values and decision-making using statistical tests
6. Confidence Intervals including estimation of population parameters with margin of error and confidence level interpretation

Module 4: Mathematics for Machine Learning

1. Linear Algebra Basics including foundational concepts, matrix operations and their role in data transformations
2. Vectors & Matrices covering vector representation, matrix multiplication and dimensional data handling techniques
3. Calculus Fundamentals including limits, functions and continuous change concepts for mathematical modeling
4. Derivatives focusing on rate of change, slope interpretation and optimization problem understanding
5. Gradient Descent covering iterative optimization techniques for minimizing errors in machine learning models
6. Cost Functions including loss measurement, error calculation and model performance evaluation methods

Module 5: Data Cleaning & Preprocessing

1. Missing Value Handling including identification, removal and imputation techniques for incomplete data
2. Outlier Detection covering methods to identify anomalies using statistical and visualization techniques
3. Data Transformation including normalization, log transformations and data formatting for improved analysis
4. Encoding techniques such as Label Encoding and One-Hot Encoding for converting categorical data into numerical format
5. Feature Scaling covering standardization and normalization methods to ensure consistent model performance
6. Data Pipelines including automated workflows for data preprocessing, transformation and model-ready dataset creation

Module 6: Exploratory Data Analysis (EDA)

1. Data Profiling including dataset overview, structure understanding and summary statistics generation
2. Univariate Analysis covering single variable exploration using distributions, central tendency and spread measures
3. Bivariate Analysis including relationship analysis between variables using comparisons and visual techniques
4. Correlation Analysis covering strength and direction of relationships using correlation coefficients and heatmaps
5. Outlier Analysis including detection of anomalies using statistical methods and visualization techniques
6. Insight Generation focusing on extracting meaningful patterns, trends and actionable business insights from data

Module 7: Data Visualization

1. Matplotlib Basics including plot creation, customization and visualization of data trends
2. Seaborn Visualization covering advanced statistical plots and aesthetically enhanced visual representations
3. Advanced Charts including heatmaps, pair plots, box plots and multi-variable visualizations
4. Storytelling with Data focusing on communicating insights effectively using visuals and narratives
5. Dashboard Basics including layout design, combining visuals and interactive reporting concepts
6. Best Practices covering chart selection, clarity, labeling and effective data presentation techniques

Module 8: Machine Learning Fundamentals

1. Introduction to ML including basic concepts, workflow and real-world applications
2. Types of ML covering supervised, unsupervised and reinforcement learning approaches
3. Train/Test Split including data partitioning techniques for model training and evaluation
4. Overfitting vs Underfitting covering model generalization issues and performance trade-offs
5. Bias-Variance Tradeoff including model complexity balancing and error minimization concepts
6. Evaluation Metrics covering accuracy, precision, recall, F1-score and performance measurement

Module 9: Supervised Learning

1. Linear Regression including prediction modeling, relationship analysis and continuous output estimation
2. Logistic Regression covering classification techniques and probability-based predictions
3. Decision Trees including rule-based modeling, splitting criteria and interpretability concepts
4. Random Forest covering ensemble learning using multiple trees for improved accuracy
5. KNN Algorithm including instance-based learning using distance metrics and nearest neighbors
6. Model Evaluation including performance comparison using classification and regression metrics

Module 10: Unsupervised Learning

1. K-Means Clustering including grouping of data points based on similarity and centroid optimization
2. Hierarchical Clustering covering tree-based clustering methods and dendrogram analysis
3. PCA including dimensionality reduction techniques and variance preservation
4. Dimensionality Reduction covering feature reduction methods for improved efficiency and visualization
5. Anomaly Detection including identification of unusual patterns and rare events in datasets
6. Use Cases covering real-world applications of clustering and unsupervised techniques

Module 11: Advanced ML & Feature Engineering

1. Feature Engineering Techniques including feature creation, transformation and domain-based enhancements
2. Feature Selection covering methods to select relevant features and reduce model complexity
3. Hyperparameter Tuning including optimization of model parameters for better performance
4. Grid Search and Random Search covering systematic and random approaches for parameter tuning
5. Ensemble Methods including boosting, bagging and stacking techniques for improved predictions
6. XGBoost and LightGBM covering advanced gradient boosting algorithms for high-performance models

Module 12: Capstone Project

1. Data Collection including sourcing and gathering structured and unstructured datasets
2. Data Cleaning covering preprocessing, missing value handling and data quality improvement
3. Exploratory Data Analysis including pattern discovery, visualization and insight extraction
4. Feature Engineering including transformation and selection of meaningful features
5. Model Building covering selection, training and optimization of machine learning models
6. Evaluation & Insights including performance analysis and deriving actionable business outcomes



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Tools List



matplotlib



seaborn

Power BI



XGB



Certification after Completion



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